

Jackson Rubiano

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Education

Bucknell University

Expected May 2027

Bachelor of Science in Computer Science & Engineering

3.72 GPA, 3.91 Engineering GPA, Dean's List

Coursework: Data Science, Applied Machine Learning, Systems Programming, Algorithm Design, Computer Networks

Work Experience

KALMUS Color Toolkit | Bucknell University

August 2024 – Present

Software Developer

- Enhanced and maintained a Python program for film color analysis using NumPy, SciPy, and OpenCV
- Developed a SQL database to store information about film analyses
- Developed a website to allow dynamic film upload and analysis retrieval
- Interfaced with public APIs to retrieve film metadata

The Japanese Paper Film Project | Bucknell University

June 2023 – Present

Software Developer

- Designed a C++ tool using OpenCV and the Blackmagic RAW SDK to extract unique frames from damaged film roll scans, reducing turnaround from 10-15+ hours to 2-4 hours
- Developed a Lua plugin for DaVinci Resolve, allowing film editors to extract frames from selected films
- Created robust filtering systems to account for varying quality, style, and inconsistent recording conditions
- Prototyped and evaluated ViT and ResNet50-based models for frame selection, comparing performance against classical computer-vision filtering
- Presented project at research symposiums and film screenings

Projects and Activities

VR Minecraft | Bucknell University

April 2025 – May 2025

Developer

- Collaborated in an Agile team to create a Minecraft-like VR game using WebGPU, WebXR, and JavaScript
- Implemented AABB collision detection and resolution and enjoyed optimizing game to achieve real-time interactivity
- Addressed difficulties integrating WebGPU for VR while maintaining near real-time rendering

ALL Blood Cell Classification | Bucknell University

March 2025 – May 2025

Developer, Author

- Built a homebrew CNN model and ResNet50 transfer learning model for classifying Acute Lymphoblastic Leukemia (ALL) cell types from peripheral blood smear (PBS) images
- Performed exploratory data analysis on PBS images to identify class imbalances and segmentation artifacts to guide model architecture and preprocessing decisions
- Implemented image segmentation pipeline to isolate relevant image regions before training models
- Interpreted model performance using confusion matrices, class-specific error analysis, and epoch vs. accuracy curves
- Documented findings and experimental methodologies in a research-style report
- Transfer learning model achieved 98% validation accuracy vs. 95% for homegrown CNN

Roguelite Game | Bucknell University

November 2023 – December 2023

Scrum Master, Developer

- Built a roguelite game in Java using JavaFX with a small team following Agile methodologies
- Designed the entity-component-system architecture and enjoyed writing core system modules
- Wrote unit/integration tests using JUnit to ensure proper functionality
- Organized daily Scrum meetings, worked to resolve obstacles, and maintained team workflow

Technical Skills

Languages C++, Objective C++, C#, Python, Java, JavaScript, TypeScript, WGSL

Technologies ResNet50, ViT, Git, Unity, Godot, ffmpeg, Blackmagic RAW SDK, .NET, SQL

Libraries Tensorflow, PyTorch, Pandas, Seaborn, OpenCV, NumPy, SciPy, Scikit-Learn, Matplotlib, JavaFX, React, OR-Tools

Testing Pytest, JUnit, GJUnit